

Hardware Prediction using Hyperdimensional Computing

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Background and Motivation

Context. Hardware predictors improve performance by speculating about future program execution in hardware.

Challenges. Predictors must be accurate and efficient to be effective. Inaccuracy results in numerous mispredictions, and if a predictor is slower than instruction execution, its predictions are futile.

Other Work. Some heavyweight hardware predictors use ML to achieve high accuracies, sacrificing performance. Some lightweight hardware predictors use saturating counters to achieve high performance, sacrificing accuracy.

Goal. We seek to understand if **hyperdimensional computing (HDC)** – a resource-efficient compute paradigm – can be used to construct an accurate and efficient hardware predictor.

Research Objective

Use HDC to perform online branch prediction – the simplest hardware prediction task – as a proof-of-concept by iterating over and encoding the history of decisions at a given program counter (PC).

Prediction Task / Branch Prediction. At branch instructions, branch predictors guess whether the branch will be taken (**T**) or not taken (**NT**) and begin executing the appropriate instruction.

- **Correct:** Program execution continues as normal.
- **Incorrect:** The CPU must dump all currently executing instructions and start over at the right one.

HDC

HDC is an error resilient associative computing model that attempts to imitate the function of the human brain.

Structure. Begin by assigning large ($N \geq 10000$) randomized binary vectors to the possible decisions (**T** and **NT**). Information is encoded in the similarity between vectors, which is calculated using the Hamming distance metric.

Operations. **Bundling** ($\oplus$) is analogous to addition and produces an output vector similar to each of its input vectors. **Binding** ($\otimes$) is analogous to multiplication. It is self-inverting and distributes over bundling. **Permutation** is a rotational bit shift and is applied along with binding to encode position.

Performing Branch Prediction using HDC

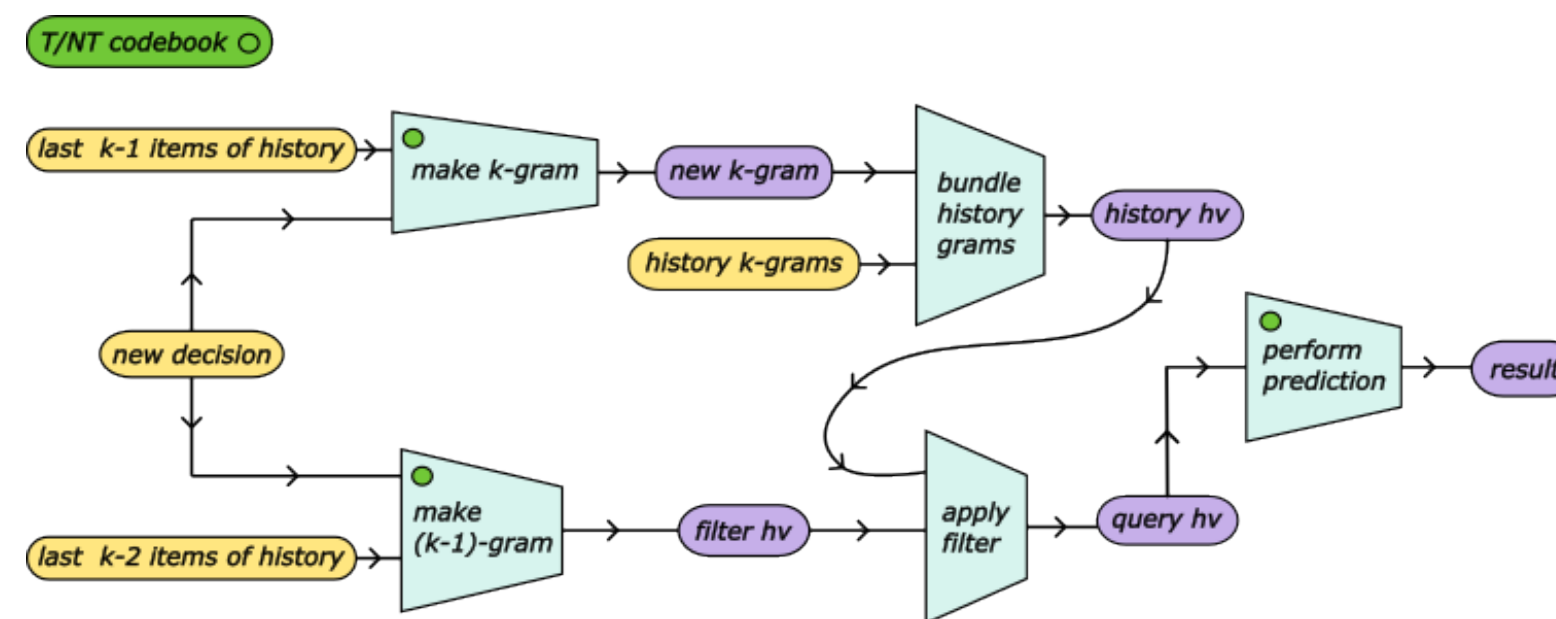


Figure 1. An overview of the encoding process

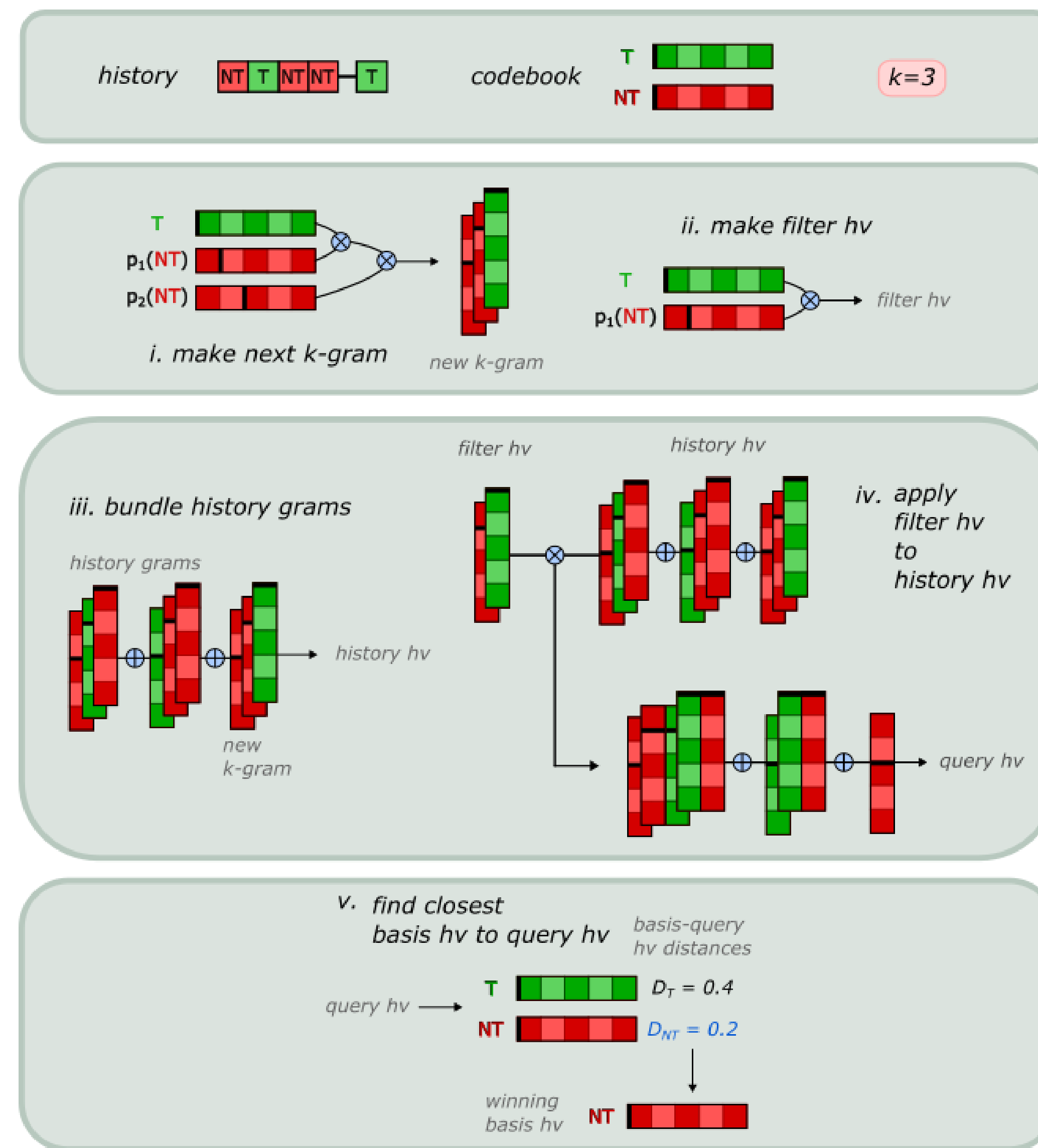
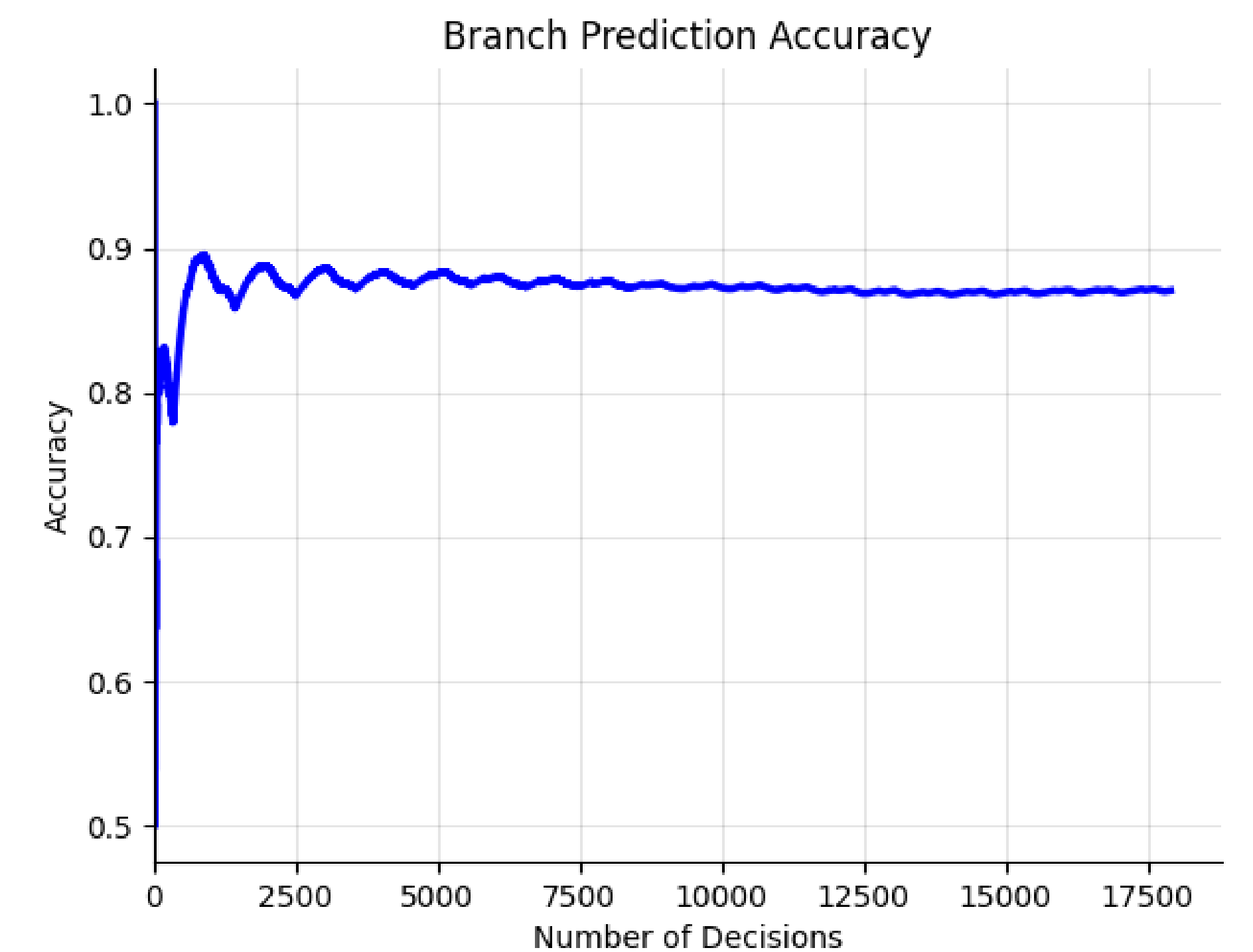


Figure 2. A detailed view of the encoding process

Preliminary Results

The predictor which we take to be our baseline is implemented using a two-bit saturating counter and has an accuracy of about 72%.

The HDC branch predictor delivered an accuracy of about 87% throughout the life of one trace.



Results were consistent. Any variations arose from the seeding of the RNG used to create the basis hypervectors and from different decision patterns at different PCs.

Conclusion and Next Steps

We will focus on improving the accuracy and efficiency of the HDC branch predictor using other information from the trace sets. We will use HDC to construct other kinds of hardware predictors.

References

- [1] David A. Patterson John L. Hennessy. *Computer Architecture A Quantitative Approach*. Elsevier, 2012.
- [2] Pentti Kanerva. *Sparse Distributed Memory*. Massachusetts Institute of Technology, 1988.